



Quantum Future Society Vision

~A Vision for the Future Society We Should Aim for Through Quantum Technology and Strategies for Its Realization~

April 22, 2022

Integrated Innovation Strategy Promotion Council

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1. Introduction

<Quantum Technology Innovation Strategy Initiatives>

In January 2020, the government formulated the Quantum Technology Innovation Strategy. To establish a hub for integrated industry-academia-government collaboration spanning fundamental research, technology demonstration, open innovation, intellectual property management, and human resource development in quantum technology, eight Quantum Innovation Hubs (QIH) were established in February 2021. Furthermore, in July of the same year, the Quantum Innovation Initiative Council (QII Council), based at the University of Tokyo, facilitated the first-ever introduction of an actual commercial quantum computer from overseas into Japan.

In the industrial sector, the Quantum Technology-based New Industry Creation Council (Q-STAR) was established in September of the same year, advancing the development of a new framework to systematically promote the research, development, and industrialization of quantum technologies.

Furthermore, concurrent with the formulation of this strategy, a forward-looking technology roadmap was also developed. Based on this roadmap, research and development is steadily progressing in various fields, including quantum computing and quantum simulation, quantum measurement and sensing, and quantum communication and cryptography.

Furthermore, international joint research between the Quantum Innovation Hub (QIH) and overseas research centers in Europe, the United States, and elsewhere is progressing steadily. This includes the hosting of the "International Symposium on Quantum Science and Technology Innovation

(Quantum Innovation 2021).

<Environmental Changes Since Formulating the Quantum Technology Innovation Strategy>

Based on the formulation of the Quantum Technology Innovation Strategy in January 2020, Japan has steadily advanced its research and development efforts. However, since the strategy's formulation, the environment surrounding quantum technology has undergone rapid and significant changes.

Overseas, private companies are leading the charge, accelerating efforts to research, develop, and commercialize quantum computers with ambitious goals. Furthermore, technologies and services integrating quantum computing with conventional (classical) computing systems are also developing rapidly. Amidst this intensifying international competition, Japan is expected to fundamentally accelerate and strengthen its quantum computer R&D, while industry, academia, and government work together to undertake full-scale, strategic efforts aimed at enhancing industrial competitiveness.

Furthermore, the COVID-19 pandemic has accelerated the digital transformation (DX) of the entire socio-economic system. Moving forward, efforts toward Society 5.0—a vision of a highly integrated cyber-physical space—are expected to advance further. Consequently, as data volumes and communication traffic explode, and security demands intensify, the importance of quantum technology—with its superior computational power and confidentiality—is growing significantly.

Furthermore, in response to the worsening climate change crisis, efforts to achieve a carbon-neutral society are gaining full momentum. Quantum technology is expected to significantly contribute to realizing this carbon-neutral society by enabling dramatic improvements in future computing performance and facilitating decarbonization through the development of unprecedented environmental materials.

Achieving large-scale quantum computers and their practical application faces numerous technical challenges, such as generating stable qubits, requiring various breakthrough technologies. In recent years, innovations in computer science, including AI, and cutting-edge research in quantum information science have made significant progress.

Remarkable progress has been made in foundational technologies for controlling and elucidating these physical phenomena, including the development of sophisticated quantum state control techniques utilizing laser technology. It is crucial to advance strategic research and development toward future breakthrough technologies while mobilizing knowledge from numerous academic fields and vigorously promoting broad-based fundamental research and human resource development, including young researchers.

In recent years, the importance of economic security has grown significantly, driven not only by the global semiconductor shortage triggered by the COVID-19 pandemic but also by the emergence of geopolitical challenges. Quantum technology is also critically important for economic security. It is vital for nations to possess advanced quantum technology domestically, including acquiring cutting-edge technologies to secure technical superiority and ensuring supply chains for critical components and materials. Furthermore, stable and continuous human resource development for this purpose is essential. Additionally, safeguarding important data and information systems that form the foundation of industry and society, including technical information, from cyberattacks and other threats is crucial.

Responding to Environmental Changes

Since the formulation of the Quantum Technology Innovation Strategy in January 2020, expectations for the role of quantum technology in addressing rapidly changing socioeconomic environments have grown. This includes intensifying international competition in quantum computing, the rapid advancement of DX driven by the COVID-19 pandemic, and efforts toward realizing a carbon-neutral society.

Furthermore, as the importance of economic security continues to grow, quantum technology is also critically important for economic security. It is vital to possess advanced technologies domestically and cultivate human resources.

Therefore, accelerating and strengthening efforts in quantum technology R&D, societal implementation, industrialization, and other areas more than ever before is an urgent task. This will ensure international competitiveness while creating growth opportunities for Japan's industries and addressing societal challenges.

The rapid advancement and widespread adoption of AI are driving the sophistication and expansion of data utilization. Quantum technology will serve as a catalyst to further accelerate this trend, bringing about a dramatic leap in future computing performance. Furthermore, quantum technology is expected to deliver significant improvements in sensing and communication capabilities, enabling vastly more sensitive sensing and highly secure communications compared to conventional methods.

Quantum technology, which will achieve such dramatic improvements in future computing, sensing, and communication capabilities, holds promise across diverse fields including drug discovery and healthcare, materials, finance, energy, lifestyle services, transportation, logistics, manufacturing, and safety and security.

Quantum technology can only create value for the socio-economic system when applied across various fields. Therefore, when promoting research and development, social implementation, industrialization, and other initiatives related to quantum technology, it is crucial to adopt a comprehensive perspective that integrates quantum technology into the entire socio-economic system.

Furthermore, quantum technology is closely intertwined with conventional (classical) technology systems in fields such as computer science (including AI and advanced simulation), information and communication technologies (including 5G/Beyond 5G), semiconductors, and measurement/sensing technologies. It is vital to advance initiatives by integrating and unifying quantum technology with these systems.

Against this backdrop, this vision is formulated as a new strategy for the future society, aiming to realize a transformation of society as a whole. It will be developed in coordination with strategies such as the AI Strategy, the Beyond 5G Promotion Strategy, and the Semiconductor Strategy, while also considering collaboration with like-minded nations. Industry, academia, and government will work together to utilize quantum technology to create growth opportunities for Japan's industries, such as a productivity revolution, and to solve societal challenges like achieving carbon neutrality.

<Relationship between this Vision and the Quantum Technology Innovation Strategy>

<Relationship between this Vision and the Quantum Technology Innovation Strategy>

This vision is formulated as a "strategy for societal transformation through quantum technology"—a future societal vision and the strategy for achieving it—aimed at leveraging quantum technology to create growth opportunities for Japan's industries, solve societal challenges, and realize transformation across society as a whole. This contrasts with the "Quantum Technology Innovation Strategy" (essentially a "quantum technology R&D strategy"), which focuses primarily on quantum technology R&D and related fields. Therefore, the government will advance its initiatives through the dual pillars of this Vision (Strategy for Social Transformation through Quantum Technology) and the "Quantum Technology Innovation Strategy" (Strategy for Quantum Technology R&D).

Therefore, moving forward, we will continue to advance research and development efforts based on the "Quantum Technology Innovation Strategy" formulated in January 2020 (with partial revisions to the roadmap). Furthermore, guided by this vision, we will promote initiatives to utilize quantum technology for creating growth opportunities for Japan's industries—such as a productivity revolution—and solving societal challenges like achieving carbon neutrality. We will advance efforts to realize the transformation of society as a whole, with an eye toward the future.

This vision was formulated with an eye toward the future society envisioned through quantum technology and the desired state to be achieved by 2030. However, as the environment surrounding quantum technology is expected to continue changing rapidly, we will continuously adapt to future environmental changes, including reviewing this vision.

2. Changes in the Environment Surrounding Quantum Technology

(1) Intensifying International Competition in the Quantum Industry

Since the formulation of the Quantum Technology Innovation Strategy in January 2020, overseas players, primarily private companies such as Google, have accelerated their efforts in quantum computer research, development, and commercialization, setting ambitious goals.

¹. Furthermore, technologies and services integrating quantum computing with conventional (classical) computing systems are also developing rapidly

Furthermore, overseas, private companies are acquiring entire university research laboratories and commercializing them, creating cases where basic research directly leads to commercialization in a single leap. Furthermore, university-spinoff ventures based on university fundamental research are also growing rapidly, backed by massive investments from venture capital, leading to IPOs and acquisitions by large corporations. Compared to the traditional linear model of fundamental-applied-practical research, the gap between fundamental research and commercialization is narrowing. Globally, efforts to realize the practical application of quantum computers are accelerating.

Amid intensifying global competition, Japan must fundamentally accelerate and strengthen its quantum computer research and development. Furthermore, it is expected that industry, academia, and government will unite to pursue full-scale, strategic efforts to enhance industrial competitiveness, while also promoting quantum industries (hardware and software vendors) and related fields utilizing quantum technology (drug discovery/medical care, materials, finance, energy, lifestyle services, transportation, logistics, manufacturing, safety, and security, etc.) that utilize quantum technology. To achieve this, industry, academia, and government must collaborate more closely to accelerate efforts in R&D, societal implementation, and industrialization, while also vigorously promoting the creation of the underlying business environment.

Furthermore, in the field of quantum communication and cryptography, overseas efforts are accelerating to establish test beds for long-distance quantum cryptographic communication utilizing terrestrial communication networks, satellites, and other space assets, and to conduct demonstration tests. Furthermore, research and development toward realizing quantum internet beyond this stage is also intensifying. Amidst this intensifying international competition in the quantum communication and cryptography field, Japan must also accelerate efforts to strengthen its international competitiveness. This requires industry, academia, and government working together, integrating with existing information and communication systems, while also taking into account the Beyond 5G Promotion Strategy.

(2) Rapid Advancement of DX Due to the COVID-19 Pandemic

Since the formulation of the Quantum Technology Innovation Strategy in January 2020, the COVID-19 pandemic has accelerated the advancement of digital transformation (DX) across all socioeconomic activities—including daily life, healthcare, education, entertainment, and transportation—as seen in DX . Moving forward, efforts toward Society 5.0, which highly integrates cyberspace and physical space, are expected to progress further.

In the post-pandemic society, new lifestyles like telework will become firmly established. Concurrently, 5G/

¹ Google (US) announced 1,000 logical qubits by 2029 (May 2021 announcement) IonQ (US) announced 1,024 logical qubits by 2028 (December 2020 announcement) Japan (Moonshot R&D Program) January 2020 announcement) targets dozens to 100 logical qubits by 2030 (accelerated schedule)

Beyond- 5G , along with technologies like IoT and edge computing, are advancing. This is expected to further accelerate the shift toward DX and Society 5.0 .

In a society advancing toward DX and Society 5.0, the volume of data handled and network traffic will increase exponentially. Furthermore, as security assurance becomes essential, relying solely on conventional (classical) technology systems may prove insufficient. Therefore, leveraging quantum computers and quantum communication/cryptography technologies—which excel in computational power and confidentiality—will become indispensable.

(3) Contributing to Carbon Neutrality and SDGs

In response to the worsening climate change crisis, countries worldwide are accelerating environmental initiatives such as carbon neutrality and beyond zero (negative emissions). Japan aims to reduce greenhouse gas emissions by 46% by fiscal year 2030 (compared to fiscal year 2013 levels). Achieving this goal requires fundamental transformation of socioeconomic systems toward decarbonization across sectors including industry, residential, and transportation.

Furthermore, to achieve the Sustainable Development Goals (SDGs: Sustainable Development Goals), faces numerous complex social challenges requiring resolution, such as healthcare, food security, and poverty. Additionally, Japan itself faces social challenges that must be overcome to build a sustainable society, including an aging population with declining birth rates and the resulting decrease in the working-age population.

Quantum technology is expected to significantly contribute to achieving a carbon-neutral society and solving complex social challenges such as the SDGs. It will enable dramatic improvements in future computing and sensing capabilities, facilitate decarbonization through the development of unprecedented environmental materials, realize a healthy and long-lived society through early disease diagnosis and treatment, and enhance productivity amid a shrinking working-age population.

(4) Advancements in Foundational Technologies Supporting Quantum Computers

Achieving large-scale quantum computers and their practical application requires numerous technological breakthroughs, including stable qubit generation and error correction. To advance research and development of these breakthrough technologies, strategic initiatives must be complemented by broad-based fundamental research grounded in theoretical approaches, which will pave the way for future breakthroughs. For instance, to generate and utilize stable qubits, it is necessary to deepen fundamental understanding of dissipative phenomena and many-body interactions within quantum systems operating in non-equilibrium states or open systems, such as those experiencing decoherence. In physics, understanding these phenomena while accounting for external environmental influences has generally been a difficult problem. However, in recent years, foundational technologies for controlling and elucidating these physical phenomena have seen remarkable development. This includes innovations in computer science, including AI; advances in quantum simulation experiments using cooled atomic systems; and the pioneering of sophisticated control technologies for quantum states utilizing cutting-edge laser technology.

It is crucial to advance strategic research and development toward breakthrough technologies essential for the future scaling and practical application of quantum computers, while also mobilizing knowledge from numerous academic fields and vigorously promoting broad-based fundamental research and human resource development, including young researchers.

(5) Economic Security of Quantum Technologies

In recent years, the importance of economic security has grown significantly, driven not only by the global semiconductor shortage triggered by the COVID-19 pandemic but also by the emergence of geopolitical challenges. Quantum technology is poised to play a foundational role in supporting future socioeconomic systems, making it critically important for economic security. It can be considered a pivotal technology at the core of future competition for global supremacy among nations.

Therefore, from an economic security perspective, it is necessary to possess advanced quantum technologies domestically, including acquiring cutting-edge technologies to secure technical superiority in quantum technology and ensuring supply chains for critical foundational components and materials. This requires the continuous and stable development and retention of human resources for this purpose. Furthermore, it is important to take measures to protect critical information systems and data that form the foundation of industry and society, including technical information, from cyberattacks and other threats.

3. Basic Approach

In light of changes in the environment surrounding quantum technology, Japan will integrate quantum technology into its socio-economic systems and leverage it to create growth opportunities for domestic industries and address societal challenges. To achieve this, industry, academia, and government will collaborate to advance initiatives based on the following three fundamental principles:

- **By integrating quantum technology into the entire socio-economic system and fusing it with conventional (classical) technology systems, (hybrid), creating growth opportunities for Japan's industries and solving societal challenges**
- **Promoting the utilization of cutting-edge quantum technologies (e.g., establishing test beds for quantum computers and quantum communication)**
- **Creation and revitalization of new industries/startup companies utilizing quantum technology**

(1) By integrating quantum technology into the entire socio-economic system and fusing it with conventional (classical) technology systems

(hybrid), creating growth opportunities for Japan's industries and solving societal challenges

Quantum technology is expected to be utilized across the entire socioeconomic system, contributing to the creation of industrial growth opportunities and the resolution of societal challenges in a wide range of fields including drug discovery and healthcare, materials, finance, energy, lifestyle services, transportation, logistics, manufacturing, and safety and security.

Therefore, when advancing research and development, social implementation, industrialization, and other initiatives related to quantum technology, it is crucial to adopt a comprehensive perspective that collaborates across various socioeconomic domains and integrates quantum technology into the entire socioeconomic system for utilization. Furthermore, it is equally important to consider quantum technology holistically, integrating it with conventional (classical) technology systems such as computer science (including AI), information and communication technologies (including 5G/Beyond 5G), measurement and sensing technologies, and semiconductors. By integrating quantum technologies and research outcomes into socio-economic systems and achieving seamless fusion and integration with conventional (classical) technology systems (), Japan can create growth opportunities for its industries, such as a productivity revolution, and realize solutions to societal challenges like achieving a carbon-neutral society and advancing the SDGs.

(2) Promoting the Utilization of Cutting-Edge Quantum Technologies (Establishing Test Beds for Quantum Computers, Communications, etc.)

For the entire socio-economic sector to utilize quantum technologies, it is crucial to create an open environment where diverse users can access quantum technologies. Furthermore, under such an environment, initiatives where users explore and create use cases by utilizing quantum technologies are also important.

Therefore, we will develop a quantum technology utilization environment, including the establishment of testbeds for cutting-edge quantum computers, quantum communication/cryptography, and quantum measurement/sensing, which diverse users from industry, academia, and government can easily access.

Furthermore, we will support initiatives where diverse users fully leverage such testbeds to explore, create, and demonstrate use cases for quantum technologies. These efforts will lead to future killer applications that significantly contribute to creating growth opportunities

for Japan's industries and solving societal challenges. Additionally, to promote the widespread adoption and expansion of quantum technologies, we will vigorously advance initiatives to disseminate information and promote best practices and

promote initiatives to disseminate and popularize information on exemplary use cases.

(3) Creation and Activation of New Industries/Startup Companies Utilizing Quantum Technology

Quantum technology is expected to be utilized across diverse industrial sectors, with both quantum computers and quantum software projected to reach massive market scales in the future². However, at present, the race remains undecided regarding the technological approach for quantum computers, the winning business model, and the key players.

The quantum field market is an emerging growth market where the outcome remains undecided. In other words, it is a market where Japan has the opportunity to win international competition and achieve its own growth, depending on its future response. Therefore, we will vigorously promote efforts to strengthen industrial competitiveness by encouraging private sector participation in the quantum field and stimulating investment.

Furthermore, in business domains focused on developing new markets, startups are expected to play a key role. These companies are resilient to environmental changes, possess flexible thinking and ideas, and excel at creating new businesses. Additionally, existing companies are also expected to generate new ventures and industries, such as creating spin-off and spin-out ventures. Corporate venture capital (CVC) is also anticipated to be utilized for investing in startups.

Therefore, industry, academia, and government will work together to vigorously promote initiatives such as developing business environments to foster and invigorate new industries and startups utilizing quantum technology, which will serve as the engine for future economic growth.

The global market size for quantum computing-related industries in 2040 is projected to be between ¥10 trillion and ¥19 trillion for hardware, and between ¥40 trillion and ¥75 trillion for software (Source: "What Happens When **“If”** Turns to **“When”** in Quantum Computing?" (BCG, July 21, 2021))

4. Future Society Vision (Future Society Image)

To promote research and development, social implementation, industrialization, and other quantum technology initiatives through integrated industry-academia-government collaboration, and to leverage quantum technology across the entire socio-economic sphere, it is crucial for stakeholders to advance these efforts while sharing a common vision for the future society. Therefore, we establish the future society vision to be pursued through quantum technology.

(1) Future Society Vision (Future Society Image) to be Pursued through Quantum Technology

In recent years, values that seek to achieve not only economic growth but also sustainability and human well-being—such as the SDGs and ESG investing—have gained greater prominence than ever before. A vision of society where the economy, environment, and society are in harmony is becoming a shared understanding in the modern era. Looking ahead to future society, such values and the initiatives based on them are expected to become increasingly important.

Therefore, we envision a future society where economy, environment, and society are in harmony as the ultimate goal to be pursued through the utilization of quantum technology. Specifically, we aim for industry, academia, and government to work together toward this vision, guided by the values of "Economic Growth and Innovation," "Harmony between People and the Environment (Sustainability)," and "Well-being."

In this context, since inappropriate use of quantum technologies—such as for cryptanalysis—could potentially harm society, efforts will be advanced by leveraging "integrated knowledge" encompassing not only natural sciences but also humanities and social sciences. This will address institutional, ethical, and societal acceptance challenges arising from the new societal application of quantum technologies, tailored to each stage of technological implementation and development.

- **Economic Growth Innovation**
- **Harmony between People and the Environment Sustainability**
- **Well-being**

(i) **Economic Growth Innovation**

Next-generation high-speed computing combining quantum computers with conventional (classical) computing will dramatically accelerate the cycle of innovation through hypothesis and verification. By leveraging quantum technology, we will achieve economic growth, including creating growth opportunities for Japan's industries such as a productivity revolution.

(ii) **Harmony between People and the Environment Sustainability**

By developing next-generation environmental materials using quantum computers and achieving carbon neutrality and a circular economy through optimal energy mixes, Japan will realize a society where people and the environment coexist in harmony and develop sustainably through quantum technology.

(iii) Well-being

Quantum cryptography enables secure and safe living; next-generation healthcare utilizing high-sensitivity quantum sensing promotes health and longevity; predictive technologies through quantum-classical hybrid computing integrate with AI; and resilient societies emerge through disaster prediction via quantum sensing and evacuation guidance systems. Quantum technology will realize a life enriched with peace of mind for all.

(2) Value Created by Quantum Technology in Future Society (Quantum Technology Application Vision)

Quantum technology has the potential to dramatically enhance future computing, sensing, and communication capabilities, creating value for society and the economy through diverse applications in drug discovery and healthcare, materials, finance, energy, lifestyle services, transportation, logistics, manufacturing, and safety and security. To realize the future society vision, potential use cases are outlined below. These represent current applications, and it is anticipated that new use cases will emerge through future discovery and technological advancement.

<Drug Discovery and Healthcare>

Drug Discovery and Pharmaceutical Manufacturing (development and production of new drugs with higher functionality and more complex structures than conventional ones), Tailored Medicine (Providing tailored medications and treatments tailored to individual patients' symptoms and constitutions), optimization of chemical reactions / Catalyst and enzyme design (enabling faster and more efficient drug discovery and manufacturing processes), high-precision diagnostics (early detection of dementia and heart disease), high-sensitivity compact MRI (aligning nuclear spins to achieve dramatically higher sensitivity and compact size with MRI), high-security communications and secure cloud (encrypted transmission of medical records and personal data, highly secure cloud services, etc.)

<Materials>

Development of new functional materials (high-performance functional materials surpassing conventional ones—energy-saving, recyclability, lightweight, toughness, etc.—and alternative materials for scarce resources), optimization of chemical reactions/design of catalysts and enzymes (efficient chemical processes contributing to a decarbonized society and circular economy, etc.)

<Finance>

Trading Strategy (Planning financial trading strategies under complex conditions, etc.), Market Forecasting (customer demand and market forecasting using quantum machine learning, accelerating derivative pricing calculations, etc.), Portfolio Optimization (optimizing held financial assets, etc.), Risk Analysis (analysis incorporating diverse risk factors through accelerated Monte Carlo simulations, etc.), Fraud Detection (building models to distinguish legitimate/fraudulent financial transactions, etc.), High-Security Communications & Secure Cloud (information related to loans and banking transactions, etc.)

<Energy>

Smart Grid (a smart system that connects and controls fluctuations on the demand and supply sides in real time, enabling real-time optimal power distribution for highly variable renewable energy sources, exploration of energy mix combinations, etc.), high-performance batteries (development of high-efficiency, high-functionality batteries using new battery materials and chemical reactions, etc.), EV energy efficiency (extending driving range per unit of energy through high-precision current sensing and control within EVs), secure infrastructure information management (SCADA), etc.

<Lifestyle Services>

High-precision weather forecasting (including high-precision forecasts at smaller regional levels), Advertising Strategy (Optimal ad placement tailored to each customer's profile and needs, etc.), Consumer Behavior (Real-time optimal product recommendations, policy combinations for environmental action, etc.), Brain-Machine Interface (**BMI**) (Non-invasive, high-sensitivity brain magnetic field measurement using quantum metrology sensing, etc.), Secure Video Conferencing (Highly confidential video conferencing using quantum cryptography, etc.)

<Transportation>

Multimodal sharing/MaaS (optimization of car sharing, on-demand ride-hailing, etc.), traffic simulation (Simulations considering complex conditions, traffic control for congestion relief, etc.), autonomous driving (route optimization, etc.), improving EV convenience (searching for charging stations and parking lots, etc.), aircraft operations network planning (aircraft allocation, crew scheduling, fuel, operational trouble response, etc.), aircraft materials (lightweight and tough new materials, etc.), wind tunnel simulations (energy-efficient aircraft design, etc.), etc.

<Logistics>

Deployment and operation planning for delivery vehicles (automobiles, ships, aircraft, drones, etc.) (maximizing load capacity/operational rates, minimizing transportation costs, etc.), route optimization (minimizing travel distance, maximizing delivery volume, minimizing transportation costs, etc.), inventory management (optimal inventory management to maximize production efficiency, efficient coordination of inventory across multiple factories, etc.), etc.

<Factory>

Smart Factory (optimization of production resources such as personnel, robots, and materials within the factory), IC manufacturing and design (e.g., complex circuit pattern combinations), secure infrastructure information management (SCADA), inventory management (optimal inventory control to maximize production efficiency, efficient coordination of inventory across multiple factories, etc.)

<Safety and Security>

High-security communications and secure cloud (highly confidential communications across all fields, high-security storage of protected data), disaster countermeasures (optimizing evacuation routes during disasters, planning personnel and supply deployment at shelters, etc.), earthquake and volcanic disaster prediction (observation within the crust using optical lattice clocks, etc.), defense applications (accelerating command and control systems)

(selection of optimal countermeasures against simultaneous multiple attacks), logistics (efficient material transport), UUV surveillance (continuous monitoring without GPS reliance), etc.

(3) Desired State by 2030 for the Future Society Vision

When promoting research and development, social implementation, industrialization, and other initiatives, it is crucial for stakeholders from industry, academia, and government to set specific numerical targets. Therefore, we will advance our efforts based on the following envisioned goals for 2030, aiming toward the future society outlined in .

- **10 million users of quantum technology in Japan**
- **Achieve a production value of 50 trillion yen through quantum technology**
- **Create quantum unicorn venture companies that pioneer future markets**

(i) Increase domestic quantum technology users to 10 million

In advanced nations, historical data indicates that internet adoption accelerates exponentially once usage rates exceed 5-10%. For domestic users of quantum technology (including quantum applications and quantum-inspired technologies, even those unaware they are using it), we aim for a similar ratio, projecting **10 million domestic users**.

To support this, we will vigorously promote the development of utilization environments, such as quantum computer testbeds, enabling diverse users to access and explore/create use cases as described earlier. We will also strongly advance initiatives to support the exploration and creation of use cases leading to killer applications, as well as efforts to disseminate and promote information about these use cases.

(ii) Achieve a production value of 50 trillion yen through quantum technology

Considering the ratio of 10 million quantum technology users to the 2030 population (119.13 million ³) and the estimated production value of industries quantum technology could contribute to (2030) of approximately 615 trillion yen ⁴, we project a production value of around 50 trillion yen. Note that this figure is based on production value.

Furthermore, the domestic value added in 2030 is projected to be approximately 1.2 trillion yen ⁵(quantum computers: 294 billion yen,

³ Source: Japan's Future Population Projections (2017 Estimates)(National Institute of Population and Social Security Research)

⁴ From the Input-Output Tables (FY2015): Manufacturing ¥302.8 trillion, Electricity ¥20.3 trillion, Commerce ¥95.5 trillion, Finance & Insurance

35.5 trillion yen, transportation 53.6 trillion yen, information and communications 50 trillion yen, healthcare 45.8 trillion yen, advertising 7.2 trillion yen. Calculated assuming a CAGR of 1% from fiscal year 2022 onwards (referencing the real GDP growth rate of +1.0% per annum in the Medium-Term Forecast for the Japanese Economy(FY2022-31)(Daiwa Institute of Research, January 24, 2022) with a real GDP growth rate of +1.0% per annum)

⁵ Source: Yano Research Institute Ltd., "Current Status and Future Outlook of the Quantum Computer Market in 2021" (September 2021)

"2022 Edition Quantum Technology Market: Current Status and Outlook"(February 2022)

Quantum Simulation: ¥48 billion, Quantum Sensing: ¥110.2 billion, Quantum Cryptography: ¥65.3 billion, Quantum Life Sciences: 32.3 billion yen, Quantum Properties 135.0 billion yen, Quantum Materials 371.0 billion yen, Quantum AI 109.8 billion yen), plus overseas earnings

(approx. ¥0.1 trillion), the total added value for both domestic and overseas markets is estimated to be approximately ¥1.3 trillion⁶.

As initiatives supporting this, the Quantum Innovation Hub (QIH), the Quantum Technology-based New Industry Creation Council (Q-STAR), the Quantum Innovation Initiative Council (QII Council) which opens pathways for actual quantum computer utilization through industry-academia collaboration, and relevant ministries and agencies. Industry, academia, and government stakeholders will collaborate more closely, intensifying full-scale and strategic efforts to enhance industrial competitiveness, including bolstering private business activities, while pursuing overseas expansion through integrated public-private efforts.

(iii) Creating quantum unicorn venture companies that pioneer future markets

In Japan, the number of unicorn companies (unlisted startup technology companies valued at over \$1 billion (approx. ¥105 billion)) has been increasing annually, reaching five companies as of December 2021: ⁽⁷⁾. Following these five companies are HIROTSU Bioscience (early cancer detection testing) (¥102.6 billion), Astroscale Holdings (space debris removal services) (¥81.8 billion), active in the emerging space business market, and ispace (development of lunar landers and rovers) (¥75.3 billion) are also rapidly growing, approaching the \$1 billion valuation threshold for unicorn ventures.

In the emerging quantum business market, we will foster unicorn companies (several in each field) across the three key quantum domains—quantum computing, quantum cryptography, and quantum measurement/sensing—by 2030 to pioneer future markets and stimulate venture capital participation.

To support this, the public and private sectors will work together to establish a comprehensive entrepreneurial environment. This includes fostering entrepreneurs, supporting research and development, facilitating connections with investors, and supplying risk capital through government-affiliated funds and other means.

⁶ Based on the ratio of domestic final demand (92.3%) and exports (7.7%) for all industries in the 2015 Input-Output Tables (Ministry of Internal Affairs and Communications), the overseas market is estimated to be approximately ¥0.1 trillion.

⁷ Preferred Networks (AI development) (¥356.1 billion), SmartNews (information gathering app)

⁷ Preferred Networks (AI development) (¥356.1 billion) SmartNews (information gathering app) (¥201.7 billion) Smart HR (cloud-based HR software) (¥173.1 billion) TBM (plastic and paper substitute materials) (¥133.6 billion) Spiber (next-generation materials) (¥131.2 billion) [Source: Nikkei Inc. "NEXT Unicorn Survey," December 2021]

5. Future Initiatives

Based on the fundamental concepts and future society vision outlined above, we will advance the following initiatives: research and development, social implementation, industrialization, and foundational efforts to foster innovation in the following quantum technology domains: quantum computers, quantum software, quantum security/networks, and quantum measurement/sensing and quantum materials.

I. Initiatives in Each Technology Area

(1) Initiatives Related to Quantum Computers

<Key Points>

- **Realization of hybrid computing systems and services combining quantum technology with conventional (classical) computing systems (including semiconductors), and a fundamental strengthening of research and development for domestically produced quantum computers comparable to those overseas**
- **Establishment of an environment to support commercialization through collaboration with domestic and international companies, including like-minded nations, and comprehensive support for industry, such as standardization assistance (establishment of a new center at AIST)**
- **Strategic R&D and promotion of fundamental research on breakthrough technologies for scaling up and practical application of quantum computers**

(i) **Realization of hybrid computing systems and services combining quantum technology with conventional (classical) computing systems (including semiconductors), and a fundamental strengthening of research and development for domestically produced quantum computers comparable to those overseas.**

- ✓ Overseas, technologies and services integrating quantum computing with conventional (classical) computing systems are rapidly advancing. For example, overseas companies are offering services that connect multiple quantum computers to the cloud, enabling the high-speed execution of hybrid quantum-classical algorithms that combine quantum computers with conventional (classical) computing systems. To respond to these developments, Japan will also actively promote research and development on hybrid computing systems and application services that integrate quantum computers with conventional (classical) computing systems like supercomputers, through industry-academia collaboration (co-creation).
- ✓ Quantum computers represent the pinnacle of cutting-edge science and technology. The scientific insights and advanced technologies gained from their development hold significant potential for widespread technological impact. Therefore, looking toward the future of Japan's science and technology, the significance of developing a domestically produced, fully functional quantum computer is immense, and we must proceed steadily. Furthermore, from the perspective that developers of quantum computers and foundational software need to quickly grasp the problems inherent in actual hardware, having access to physical machines that allow tuning of hardware and middleware is essential.

It is essential to establish a test bed equipped with an environment where actual machines can be utilized, both to achieve this goal and to establish de facto standards in core technologies. Therefore, while building a supply chain with companies from like-minded countries, we will vigorously advance the research and development of domestically produced quantum computers and steadily progress efforts to establish a test bed with broad applicability. Specifically, we will establish the first prototype in fiscal year 2022. Subsequently, based on international trends, R&D progress, and user demand for the testbed, we will steadily advance the enhancement of the testbed and necessary R&D.

- ✓ As overseas companies set ambitious goals one after another toward scaling up and practical application of quantum computers, intensifying international competition, Japan will also set ambitious goals comparable to those abroad. With an eye toward collaboration with conventional (classical) computing systems and in cooperation with like-minded nations, Japan will strategically and fundamentally strengthen and accelerate research and development efforts toward realizing future fault-tolerant, domestically produced quantum computers.

(ii) Comprehensive support for industry, including environment development for commercialization through collaboration with domestic and international companies (including partner nations) and standardization support (e.g., establishing a new center at AIST).

- ✓ For private companies to research and develop quantum computers requires long-term and massive investment. Moving forward, some point out that relying solely on the R&D resources of Japanese companies and research institutions to advance the research and development of prototyping and manufacturing technologies for large-scale chips with thousands or tens of thousands of qubits, and further, to proceed with device implementation for commercialization, is difficult and carries high risks.
- ✓ Overseas, there are exemplary cases like Belgium's IMEC (Interuniversity Microelectronics Centre) at . It possesses a world-leading research infrastructure for semiconductor manufacturing process technology and serves as a core industrial support hub for semiconductor R&D, prototyping, and manufacturing through collaborative research with domestic and international companies.
- ✓ Based on these considerations, the National Institute of Advanced Industrial Science and Technology (AIST) will collaborate with domestic and international companies, including interested nations, to provide private enterprises with opportunities to prototype, manufacture, and evaluate quantum chips and peripheral devices, and the development of new user markets, including service businesses offering access to quantum-classical hybrid computing resources, as well as commercialization. It will also provide standardization support. This initiative aims to establish a "Global Industrial Support Center (tentative name)" that comprehensively supports industry from a global perspective, with an eye toward future commercialization. Furthermore, this initiative will involve prototyping, manufacturing, and evaluating quantum computer chips and peripheral control devices through collaboration with semiconductor prototyping and manufacturing facilities, semiconductor technology, and measurement technology. We will also promote initiatives with an eye toward the mutual linkage and advancement of diverse technologies such as quantum technology, semiconductors, and energy. This includes introducing outstanding components and electronics technologies possessed by private companies (including SMEs such as domestic and international venture companies) and research institutions, and integrating efforts from other fields such as materials informatics and energy management to create user markets.

(iii) Strategic research and development of breakthrough technologies and promotion of fundamental research for the large-scale development and practical application of quantum computers

- ✓ Achieving large-scale quantum computers and their practical application requires numerous technological breakthroughs, including stable qubit generation and error correction. Furthermore, beyond the currently promising quantum computing technologies—such as superconducting, ion trap, photonic, and silicon qubits—breakthrough technologies involving entirely novel approaches are also anticipated. To achieve these breakthrough technologies, alongside strategic R&D, broad-based fundamental research grounded in theoretical approaches—such as exploring the principles of quantum systems in non-equilibrium states and open systems—is crucial as it can lead to breakthrough technologies.
- ✓ Therefore, we will advance strategic research and development toward breakthrough technologies essential for the future scaling and practical application of quantum computers. Simultaneously, we will mobilize knowledge from numerous academic fields and vigorously promote broad-based fundamental research and human resource development, including young researchers.
- ✓ Furthermore, optical technologies such as lasers have brought significant innovation to advanced control of quantum states, high-precision microfabrication, and materials development. They serve as an indispensable critical technological foundation that comprehensively supports quantum technology, including research and development, practical application of quantum measurement and sensing, and the manufacturing of components, materials (such as quantum chips and quantum repeaters), and related semiconductors. This is essential for quantum computers, quantum networks, quantum measurement and sensing, as well as the manufacturing of components, materials (such as quantum chips and quantum repeaters), and related semiconductors. Therefore, we will enhance and strengthen broad-based fundamental research, including strategic R&D and human resource development.

(iv) Other (Points to Note)

- ✓ In pursuing quantum computer R&D, it is crucial to adopt a strategic and flexible approach to accelerate, enhance, and adjust research and development, taking into account international trends.
- ✓ When formulating future business strategies for quantum computing, it is crucial to also consider creating environments that provide access to quantum computers. Furthermore, it is expected that business strategies will be developed with a balanced approach, encompassing not only vertical integration but also horizontal division of labor that leverages Japan's strengths while anticipating collaboration with overseas partners.
- ✓ In the research and development of quantum computers, it is expected that in the short term, the provision of services utilizing quantum annealing (Ising machines) will be steadily advanced. Furthermore, by utilizing testbeds such as NISQ (Noisy-Intermediate-Scale Quantum Computer) (gate-based machines), opportunities for their use will be expanded. The insights gained from this will then be leveraged for the future provision of full-scale services using large-scale gate-based quantum computers.
- ✓ In pursuing research and development of quantum computers, it is crucial to adopt a long-term perspective, foster exchange and collaboration between corporate and academic talent, establish an open innovation framework involving numerous companies, develop environments for nurturing professionals and other personnel, encourage participation from researchers and engineers in peripheral fields beyond quantum technology, and further promote international talent exchange.
- ✓ For the utilization of quantum technology by users, efforts centered on conventional (classical) technology—including peripheral equipment technology—are necessary for the device implementation, commercialization, and maintenance required during the operational phase.

Therefore, for the practical application and commercialization of quantum computers, it is necessary to secure commitment from private companies and establish a framework for collaboration between private companies and quantum technology experts.

- ✓ To expand and promote future investment from industry, it is also important to explore use cases and quantitatively demonstrate their economic and social impacts.

(2) Initiatives on Quantum Software

<Key Points>

- Establishing a quantum computer usage environment accessible to diverse users for exploring and creating use cases (e.g., testbed development)
- Development of software through convergence with conventional (classical) technology fields such as AI, as well as other fields like drug discovery/medicine, materials, and finance (), with an eye toward hybrid quantum-classical computing services (industry-academia co-creation)
- Fundamentally enhancing and strengthening national projects on quantum software, establishing mechanisms to discover and support outstanding ideas

(i) Establishment of a quantum computer usage environment enabling diverse users to access and explore/create use cases (e.g., establishing testbeds)

- ✓ To leverage quantum technology across the entire socio-economic landscape, we will establish utilization environments such as quantum computer testbeds. These will enable diverse users from industry, academia, and government to access the technology and explore and create use cases. In doing so, we will flexibly utilize overseas quantum computers while establishing an environment for accessing cutting-edge quantum computers.
- ✓ Furthermore, we will support initiatives where diverse users fully leverage such testbeds to explore, create, and demonstrate use cases for quantum technologies. These efforts will lead to future killer applications that significantly contribute to creating growth opportunities for Japan's industries and solving societal challenges.
- ✓ Furthermore, to enhance user literacy and promote the participation and utilization of user companies, we will vigorously advance initiatives to disseminate and promote information on exemplary use cases.

(ii) Development of software through convergence with other fields such as drug discovery/medical care, materials, finance, and conventional (classical) technology fields like AI, with an eye toward hybrid quantum-classical computing services (industry-academia co-creation).

- ✓ Quantum applications are expected to contribute to creating growth opportunities across a wide range of industries—including drug discovery and healthcare, materials, finance, energy, lifestyle services, transportation, logistics, manufacturing, and safety and security—while also addressing societal challenges such as achieving carbon neutrality. Furthermore, hybrid computing services that integrate with conventional (classical) technology systems—such as computer science (including AI) and information and communication technologies (ICT) like 5G/Beyond 5G—will

It holds significant potential to generate unprecedented scientific achievements that transcend existing physics frameworks, in addition to delivering new industrial value.

- ✓ For the research and development of quantum applications, it is essential that universities and research institutions possessing advanced expertise in quantum information science, such as quantum algorithms, closely collaborate with various companies (software vendors, user companies) to develop applications (i.e., co-creation).
- ✓ Therefore, we will fundamentally enhance both the quality and quantity of open innovation platforms where universities, research institutions, and others collaborate with industry stakeholders across multiple fields to advance research research institutions, and others will fundamentally enhance both the quality and quantity of open innovation platforms where they collaborate with stakeholders from various industries to advance research and development and societal implementation. Furthermore, under this collaborative framework, we will vigorously promote research and development of software such as application algorithms and their societal implementation, with an eye toward future hybrid computing services that integrate quantum technology with conventional (classical) technology.

(iii) Fundamental enhancement and strengthening of national projects on quantum software, mechanisms to discover and support outstanding ideas

- ✓ In quantum computer research and development, it is crucial to pursue not only hardware but also software R&D, such as application algorithms. By providing users with services that deliver customer value, we can secure both direct and indirect revenue streams. This enables a virtuous cycle of funding, where these revenues are reinvested into next-generation R&D. Furthermore, treating hardware and software as two wheels driving R&D is essential. This approach allows insights gained from software to be fed back into hardware development, mutually advancing both hardware and software sophistication.
- ✓ Currently, while national projects exist for the research and development of hardware such as quantum computers, national projects for quantum software research and development are limited in both quality and quantity. Therefore, we will work to enhance and strengthen national projects related to quantum software going forward.
- ✓ As key players in quantum software R&D, we anticipate participation from young talent with flexible and innovative ideas, startups, and individuals from other fields such as drug discovery/medicine, materials, finance, as well as conventional (classical) technology domains like AI. To open doors as widely as possible and broadly incorporate ideas from society, we will consider and implement mechanisms to discover and support outstanding ideas through initiatives like idea contests.

(iv) Other (Points to Note, etc.)

- ✓ In quantum software research and development, the necessity for actual hardware varies significantly depending on one's position, objectives, and intended use. Therefore, flexible strategies encompassing the utilization of domestic and international hardware, along with the establishment of an environment for accessing quantum computers that attracts diverse talent, are essential.
- ✓ For user companies, engaging with quantum technologies—which carry high future uncertainty—often proves challenging from an investor accountability perspective. Therefore, it is crucial to first create and accumulate small achievements (use cases). Particular emphasis should be placed on identifying fields with numerous potential users, low implementation barriers, significant market potential and impact, and areas where Japan holds competitive advantages, and conducting R&D on quantum applications from these perspectives.

is also crucial.

- ✓ When disseminating information and promoting future quantum computer use cases, it is also important to create a common language between stakeholders in quantum and non-quantum fields—such as by expressing customer journeys through illustrations and narratives—to stimulate societal demand and private investment.
- ✓ In the research and development of quantum applications, it is extremely important for private companies to have a permanent presence at universities, creating an environment where researchers can interact face-to-face. Furthermore, to leverage flexible and free thinking to generate unprecedented ideas, it is crucial to foster an environment that respects the playful spirit and ingenuity of researchers and developers.
- ✓ When conducting idea contests, mechanisms such as disclosing open issues per use case and offering prize money can be effective. Since Proof of Concept (PoC) for the use cases themselves, aimed at customer value, also constitutes valuable ideas, mechanisms to generate and discover these ideas are also effective.

(3) Initiatives on Quantum Security and Quantum Networks

<Key Points>

- Expand and **enhance quantum cryptography communication testbeds and utilization demonstrations. Realize comprehensive security integrating quantum technologies with conventional (classical) technologies, including post-quantum cryptography.**
- Support measures such as evaluation and certification systems to promote the adoption of quantum cryptography communication technology
- Launch of a National Project for Research and Development of Quantum Internet Enabling Communication While Maintaining Quantum States

(i) **Expansion and enhancement of quantum cryptography communication test beds and application demonstrations; realization of comprehensive security integrating quantum technology with conventional (classical) technology, including post-quantum cryptography**

- ✓ Internationally, competition in quantum cryptography is intensifying, with accelerated efforts to establish test beds for long-distance quantum cryptographic communications utilizing terrestrial communication networks, satellites, and other space assets, and to conduct demonstration tests. From an economic security perspective, it is essential for Japan to secure advanced quantum cryptography technologies and accelerate their societal implementation.
- ✓ Therefore, we will expand and enhance the open testbed for quantum cryptography communication networks, broaden user participation in utilization demonstrations to advance commercialization, and explore future network development connecting cities, including the utilization of space assets such as satellites.
- ✓ Furthermore, public and private sector users, vendors, service providers, telecommunications and cloud operators, and others will closely collaborate and consult, accumulating practical demonstration results. While advancing standardization in response to required security requirements, they will promote the creation and accumulation of use cases for comprehensive quantum security technologies that integrate quantum and classical systems, including the utilization of post-quantum cryptography and secret sharing technology.

(ii) Support measures such as evaluation and certification systems to promote the adoption of quantum cryptography communication technology

- ✓ Within Japan, some private companies have already launched services utilizing quantum security technology, and it is anticipated that multiple providers will offer such services in the future.
- ✓ At this stage, when users consider adopting such services, it is crucial that they can understand and evaluate, using objective and unified metrics, the extent to which these services ensure security and whether they meet user needs.
- ✓ Therefore, to promote the adoption of services utilizing quantum security technology, while continuing standardization efforts, we will examine necessary support for initiatives such as introducing evaluation and certification systems for quantum cryptography communication devices, which are advancing toward practical implementation, as we work toward building next-generation industrial security.

(iii) Launch of a National Project for Research and Development of Quantum Internet Enabling Communication While Maintaining Quantum States

- ✓ The quantum internet, the ultimate form of quantum networks enabling communication while maintaining quantum states, is anticipated as a foundational communication technology for various quantum applications. These include secure communication, scaling up quantum bit counts and enabling distributed computing through connecting multiple quantum computers, and networked quantum sensors.
- ✓ In recent years, overseas research and development of key technologies for realizing quantum internet has intensified. This includes quantum repeaters enabling the relay of quantum states, quantum memory capable of storing quantum states, and the establishment of testbeds for demonstrating these technologies.
- ✓ In Japan, in coordination with efforts to realize next-generation communication systems such as Beyond 5G and 6G, we will commence research and development on the elemental technologies for quantum internet. This will be based on a roadmap for quantum internet development, aiming to establish the technological foundation for scaling future quantum computers and advancing quantum cryptographic communication as a quantum communication infrastructure.

(iv) Other (Points to Note, etc.)

- ✓ To gain functionality and meaning through universal use. Therefore, it is crucial for the public and private sectors to collaborate, including integration with information and communication systems such as 5G/Beyond 5G and optoelectronic convergence technologies. This collaboration should advance initiatives with a vision for the future grand design of security networks and the contribution and positioning of quantum technologies.
- ✓ For commercialization, the user perspective is more important than the technical perspective, and providing services that allow users to easily utilize quantum cryptography is crucial. Therefore, a service mechanism is expected where new technologies are gradually applied to quantum key distribution (QKD) networks without changing the user interface.
- ✓ Security measures tend to be postponed without enforceability, making it difficult for vendor companies to invest and for user companies to adopt. Therefore, in the future, mechanisms such as guidelines and institutional frameworks to enhance incentives for using quantum security are also anticipated.

and institutional frameworks to boost incentives for adopting quantum security.

- ✓ To promote the adoption of quantum cryptography, it is crucial to establish a framework enabling stakeholders—including users, academia, and engineers—to foster a shared understanding and collaborate on addressing challenges.
- ✓ While steadily advancing the practical demonstration, advancement, and overseas deployment of quantum key distribution (QKD) network technology—an area where Japan currently holds strengths—it is equally important to maintain a forward-looking perspective. This involves concurrently pursuing research and development on next-generation quantum network/quantum internet technologies, striving to maintain and enhance Japan's international competitiveness in quantum security and quantum network technologies.

(4) Initiatives in Quantum Measurement/Sensing and Quantum Materials

<Key Points>

- **Expanding application fields and use cases for quantum measurement and sensing technologies; establishing utilization environments (e.g., test bed development); enhancing and strengthening the technological foundations supporting their utilization**
- **Identifying companies (users/vendors) with future business strategies and supporting their commercialization**
- **Establishment of R&D and supply infrastructure for quantum materials demonstrating world-leading quantum functionality**

(i) **Expanding application fields and use cases for quantum measurement and sensing technologies, establishing utilization environments (including testbeds), and enhancing the technological foundation supporting their utilization**

- ✓ Quantum measurement and sensing technologies enable significantly higher precision and sensitivity than conventional methods, and are expected to find broad applications across fields such as life sciences and medicine, mobility, energy, and disaster prevention. Furthermore, they are anticipated to evolve in combination with AI, big data, IoT, and other technologies.
- ✓ On the other hand, given that the application fields and use cases currently envisioned for such latent needs remain limited, it is crucial to expand into further application fields and create new use cases. Furthermore, enhancing and strengthening the advanced technological infrastructure that supports such utilization is also essential.
- ✓ Therefore, we will advance the development of an environment enabling users from industry and academia across many fields to understand and utilize quantum measurement and sensing technologies, while also supporting the creation of use cases that will become future killer applications and their practical verification. Furthermore, to address emerging utilization needs and enhance international competitiveness, we will advance fundamental research supporting technological sophistication. We will also strengthen the technological foundation through integrated R&D spanning conventional (classical) technologies— , AI/Big Data, IoT ()—and diverse fields including life sciences/medicine, mobility, energy, and disaster prevention, while fostering and securing talent.

(ii) Identifying and supporting the commercialization of companies (users/vendors) with an eye toward future business strategies.

- ✓ Quantum measurement and sensing hold significant potential for various applications, yet the high technical barriers make it difficult for many companies to enter this field.
- ✓ Furthermore, while leveraging the unique characteristics of quantum measurement and sensing technology could unlock potential demand for new applications, there is currently insufficient opportunity for needs and seeds to meet. Additionally, awareness among companies and other entities tends to be low, resulting in an inability to fully identify potential user companies.
- ✓ Therefore, universities and research institutions will actively provide technical support and information to potential vendor and user companies interested in quantum measurement and sensing technologies. They will also establish mechanisms to match user companies (needs) with universities and research institutions (seeds). Furthermore, mechanisms to discover and support new ideas, such as through idea contests, will be explored.

(iii) Establishment of an R&D and Supply Infrastructure for Quantum Materials Capable of World-Leading Quantum Functionality

- ✓ Quantum computers, quantum and measurement sensing, quantum networks and quantum internet, and other next-generation devices utilizing quantum materials require quantum materials capable of demonstrating cutting-edge quantum functionality for their research, development, practical application, and fundamental research.
- ✓ Furthermore, from the perspective of economic security, it is critically important for countries to possess domestic manufacturing technologies for cutting-edge quantum materials and to establish stable supply bases.
- ✓ Therefore, we will strengthen research and development on the creation and advancement of quantum functions using high-precision quantum beam technology. Through this creation and advancement of quantum functions, we will establish a "Quantum Function Creation Center (tentative name)" that will serve as the core for developing world-leading research and development on quantum materials capable of demonstrating high-performance quantum functions, as well as for establishing a supply infrastructure for advanced quantum materials that leads the world and ensuring their stable supply.

(iv) Other (Points to Note, etc.)

- ✓ Quantum measurement and sensing face significant technical hurdles and tend to be less well-known among companies and other entities. Therefore, it is crucial to secure interpreters (translators/explanators) who can effectively digest and articulate how these technologies can be applied, or personnel who can become business leaders.
- ✓ To identify killer applications and establish them as viable businesses, it is considered effective for vendor companies and user companies to collaborate, with user companies planning and leading the business initiatives.

II. Foundational Initiatives for Innovation Creation

(1) Creation and Activation of Startups

<Key Points>

- Comprehensive startup ecosystem development including entrepreneur training, business commercialization support, investor matching, and leveraging government funds; discovery and support of new businesses through idea contests/pitch contests
- Support for startup companies in utilizing quantum computers and conducting application R&D
- Improving procurement of products and services from startups and SMEs (e.g., procurement in research projects that values factors beyond track record and price, procurement that includes technological advancement beyond mere purchasing)

(i) Comprehensive entrepreneurship ecosystem development, including entrepreneur training, business commercialization support, investor matching, and utilization of government-backed funds, alongside idea contests/pitch contests to discover and support new businesses

- ✓ In business domains pioneering new markets, startups are expected to thrive—those resilient to environmental changes, possessing flexible thinking and ideas, and adept at creating new businesses.
- ✓ To foster and invigorate quantum startups, we will establish a comprehensive entrepreneurial ecosystem. This includes: - Entrepreneur development programs - Business commercialization support - Investor matching assistance - Risk capital provision through government funds and similar sources - Research and development support - Provision of quantum computing utilization environments - Collaboration with incubation hubs
- ✓ Additionally, to foster new businesses leveraging quantum technology, we will host idea contests and pitch contests targeting startups and young talent to discover and support new business opportunities.

(ii) Support for startup companies in utilizing quantum computers and conducting application research and development

- ✓ Application businesses are knowledge-intensive and do not require large-scale investment. They often demand flexibility to customize and localize services for each customer. Therefore, this represents a promising business area for agile startups capable of flexible responses.
- ✓ However, for startups considering or already pursuing quantum applications as a business, capital resources may not always be sufficient, creating significant barriers to quantum application R&D and the use of quantum computers for this purpose.
- ✓ To support the business activities of such startups, we will provide assistance to promising startups for quantum application R&D, proof-of-concept testing, and quantum computer usage.

(iii) Procurement improvements for products and services from small and medium-sized enterprises, including startups (e.g., procurement that prioritizes factors beyond track record and price in research projects, procurement that encompasses technological advancement rather than merely purchasing goods)

- ✓ Quantum business requires long-term investment, and in small, nascent markets, market forces alone may not be sufficient to sustain viable businesses. Therefore, it is crucial to nurture startups and other small and medium-sized enterprises (SMEs) with a long-term perspective. Furthermore, for SMEs, targeting not only the domestic market but also overseas markets to secure a certain market scale can be an effective strategy to enhance business viability.
- ✓ On the other hand, in projects undertaken by national governments and research institutions, procurement of equipment and devices often prioritizes proven track records and price, frequently placing startups and other small and medium-sized enterprises at a disadvantage. Furthermore, such procurement may remain limited to mere purchasing, failing to contribute to securing or enhancing international competitiveness.
- ✓ Therefore, from the perspective of fostering small and medium-sized enterprises, including startups, the government and research institutions will strive to fulfill their role as early adopters in stimulating demand by implementing procurement practices that emphasize future growth potential alongside performance and price. Furthermore, when procuring equipment and devices from SMEs, efforts will be made strategically—not merely as purchases—with an eye toward enhancing international competitiveness and expanding into overseas markets. This includes incorporating research and development for advancement as necessary.

(iv) Other (Points to Note, etc.)

- ✓ Overseas startups often receive technical support from researchers in some capacity. Therefore, to increase the number of startups in Japan, it is crucial to simultaneously increase the number of researchers and students in the quantum field. Furthermore, comprehensive support for university-spinoff startups (covering intellectual property strategy, legal affairs, business strategy, etc.) is also vital.

(2) Strengthening the Framework of Quantum Technology Innovation Hubs

<Key Points>

- **Formation and functional enhancement of new hubs for industrial competitiveness, economic security, quantum technology utilization, and strengthening international competitiveness**
- **Fundamentally enhance the headquarters function representing Japan's quantum technology innovation hub (RIKEN)**

(i) **Formation of new hubs and functional enhancement for industrial competitiveness, economic security, quantum technology utilization, and strengthening international competitiveness**

- ✓ Regarding the Quantum Technology Innovation Hubs, eight ⁸hubs were launched in February 2021. Each hub leverages its unique strengths to advance initiatives in research and development, international collaboration, industry-academia-government partnerships, and human resource development.
- ✓ To invigorate and fully launch activities aimed at commercializing quantum technology within industry, it is necessary for Japan to strengthen its support for industry, with a view to securing international competitiveness in both hardware and software.
- ✓ Regarding hardware, as advancements and diversification in quantum bit development are anticipated, it is necessary to elevate capabilities beyond the laboratory level to an industrial technology standard and aim for domestic production of quantum computers and key technologies. To achieve this, comprehensive and seamless support for industry activities—such as prototyping, manufacturing, evaluation, and standardization of quantum chips—is essential. Furthermore, from an economic security perspective, it is also necessary to advance quantum materials essential for the research, development, and industrialization of quantum devices, and to establish and strengthen the supply chain infrastructure.
- ✓ Regarding the software aspect, as new business activities such as services integrating quantum and classical computing are expected to become more active and mature, it is necessary to establish an advanced hybrid computing environment that seamlessly connects quantum and classical systems. Furthermore, to effectively utilize quantum technology within socio-economic systems, it is also essential to support the creation of quantum-based solutions and cultivate industrial talent who will become key players in this field.
- ✓ Furthermore, to prevent Japan from falling behind in the fierce international competition in the quantum field, it is necessary to incorporate the latest academic theories and technologies from overseas industry and academia. To this end, it is also essential to strengthen international research and development, education, and other activities, while establishing internationally outstanding research hubs to enhance Japan's appeal.

⁸ Quantum Computer Development / Headquarters for All Centers (RIKEN) Quantum Computer Utilization (University of Tokyo - Corporate Consortium) Quantum Device Development (National Institute of Advanced Industrial Science and Technology) Quantum Software (Osaka University) Quantum Security (National Institute of Information and Communications Technology) Quantum Life (Bio) Japan Science and Technology Agency) Quantum Materials (National Institute for Materials Science) Quantum Sensors (Tokyo Institute of Technology)

- ✓ Based on these considerations, to enhance industrial competitiveness, economic security, quantum technology utilization, and international competitiveness, the following measures will be implemented to strengthen the framework through establishing new hubs and enhancing existing functions.

- **"Global Industrial Support Center (tentative name)" (National Institute of Advanced Industrial Science and Technology (AIST))**

In collaboration with domestic and international companies, including interested nations, we will establish a hub to provide comprehensive support to industry. This includes fostering new user markets for private companies through prototyping, manufacturing, and evaluation of quantum chips and peripheral devices; offering opportunities to utilize quantum-classical hybrid computing resources; and facilitating commercialization. We will also develop an enabling environment and provide standardization support, all with a global perspective focused on future commercialization.

- **"Quantum Function Creation Center (Provisional Name)" (Quantum Science and Technology Research Development Organization (QST))**

Through the creation and advancement of quantum functions using high-precision quantum beam technology, we will establish a world-leading research and development hub for quantum materials that demonstrate high-performance quantum functions. This hub will also serve as a base for developing a globally leading supply infrastructure for advanced quantum materials and ensuring their stable supply.

- **"Quantum Solutions Hub (tentative name)" (Tohoku University)**

Collaborating with numerous user and vendor companies, we will establish a hub to support the utilization of quantum technology, foster the research and development of valuable solutions aligned with corporate needs, build hybrid quantum-classical computing environments, and cultivate industrial talent through these efforts.

- **International Education and Research Hub (Provisional Name) (Okinawa Institute of Science and Technology Graduate University (OIST))**

This hub will bring together outstanding quantum researchers from Japan and abroad to advance cutting-edge international collaborative research. It will also provide world-class international education in quantum technology, establishing itself as a global center for leading-edge research, development, and education.

- (ii) Fundamental strengthening of the headquarters function representing Japan's quantum technology innovation hub (RIKEN)**
 - ✓ To ensure Japan does not fall behind in the fierce international competition in the quantum field, it is essential to establish a system that promptly incorporates the cutting-edge academic theories and technologies being advanced by overseas industry and academia. To achieve this, it is necessary to advance internationally advanced research, development, and education in quantum technology, both in fundamental and applied aspects, thereby enhancing Japan's appeal. Concurrently, it is essential to create internationally visible hubs that form the core of these efforts.
 - ✓ RIKEN is positioned as the headquarters center that oversees and represents all quantum technology innovation hubs in Japan (RIKEN Center for

Quantum Computing).

- ✓ As quantum technology expands into various fields, the headquarters must establish a structure capable of covering a broad range of scientific and technological domains. Therefore, we will develop and strengthen an environment where top researchers from Japan and abroad can gather to advance world-class cutting-edge research, under a framework that provides a broad overview from both scientific and technological perspectives. Specifically, we will deepen collaboration with domestic and international research institutions across industry, academia, and government to maintain and enhance Japan's overall cutting-edge scientific and technological capabilities. Furthermore, we will fundamentally strengthen the headquarters' functions, including cross-sector international collaboration encompassing industry, information dissemination, and coordination among multiple bases.
- ✓ Additionally, it will serve as a hub for providing advanced scientific computing and data utilization methods, including the realization of hybrid computing systems and services through the integration of quantum computers and conventional (classical) computing systems. Therefore, it is crucial to strengthen the research collaboration framework across related fields such as quantum computing, supercomputing, AI, and bioinformatics, concurrently with enhancing this hub.

(iii) Other (Points to Note, etc.)

- ✓ It is important for the Quantum Technology Innovation Hubs to promote internationally competitive research and development, further enhance functions such as industry-academia collaboration and international cooperation, and strengthen the infrastructure of each hub (including improving positions and benefits) to make them more attractive and appealing to domestic and international industry and researchers.
- ✓ It is also vital that the Quantum Technology Innovation Hubs leverage their respective strengths while collaborating with each other to maximize functions such as industry-academia-government collaboration and international cooperation. They should actively disseminate information to enhance their presence both domestically and internationally.

(3) Human Resource Development and Securing

<Key Points>

- **Providing education programs (including lifelong learning) to a broad range of stakeholders, including industry players utilizing private sector entities, and centralizing the provision of related information**
- **Developing talent through integration with other fields such as drug discovery/medicine, materials, finance, and conventional (classical) technology areas like AI (e.g., hybrid talent combining "●●×quantum" where quantum is a second language)**
- **Cultivating a broad base of young research talent to drive future breakthrough technologies, and fostering quantum natives through science museum exhibits, video content, etc. (e.g., creating environments where children encounter quantum concepts from an early age)**

(i) Providing educational programs to a broad audience, including industry with the involvement of private sector entities (such as continuing education), and centralizing the provision of related information.

- ✓ In recent years, there has been a growing movement, primarily among researchers at universities and research institutions, to develop and provide educational programs in the quantum field.

It is becoming increasingly active. However, given the limited resources available to universities, research institutions, and other entities providing these programs, utilizing private sector entities to implement them is considered an effective approach for expanding educational programs to a broader audience in the future.

- ✓ Therefore, we will leverage private-sector entities to provide educational programs across a broad spectrum, including industry sectors such as drug discovery and medicine, materials, finance, and conventional (classical) technology fields like AI. We will also establish mechanisms to match industry and academia talent, enabling young professionals to smoothly choose quantum fields as attractive future career paths. Looking ahead, we will pursue initiatives with an eye toward introducing certification systems that visualize and validate technical proficiency.
- ✓ Additionally, we will establish a system to provide centralized and systematic information on educational programs offered by multiple institutions.

(ii) Developing talent that integrates with other fields such as drug discovery/medicine, materials, finance, and conventional (classical) technology fields like AI (e.g., hybrid talent combining "●●×quantum" with quantum as a second language).

- ✓ Quantum technology can only create and deliver user value when integrated with other fields like drug discovery/medicine, materials, finance, and conventional (classical) technology fields. Therefore, a perspective that involves collaboration with other fields and conventional (classical) technology fields is crucial in human resource development as well.
- ✓ Furthermore, from the perspective of strengthening human resources in the quantum field and cultivating individuals who can bridge technology seeds and user needs, it is crucial to develop professionals possessing expertise in both quantum technology and other fields or conventional (classical) technology domains. In the future, when utilizing quantum technology becomes commonplace, hybrid talent—those who specialize primarily in other fields or classical technology domains while mastering quantum technology, effectively treating quantum as a second language—will also be in demand.
- ✓ To cultivate such hybrid talent, we will establish a framework providing integrated educational programs that fuse quantum fields with other disciplines and conventional (classical) technology fields. Furthermore, we will promote talent development by actively introducing research and development themes that integrate quantum fields with other disciplines and conventional (classical) technology fields within R&D projects.

(iii) Cultivating a broad base of young researchers who will drive future breakthrough technologies, and fostering quantum natives (e.g., creating environments where children are exposed to quantum concepts from an early age) through science museum exhibits, video content, and other resources.

- ✓ Achieving large-scale quantum computers and practical applications requires numerous breakthrough technologies, such as stable quantum bit generation and error correction.
- ✓ We will promote broad-based fundamental research toward future breakthrough technologies and vigorously advance human resource development, including young researchers. In doing so, we will foster environments enabling broad-based human resource development across numerous universities, research institutions, and other entities, including those outside the Quantum Technology Innovation Hubs. Furthermore, national research and development corporations responsible for policy-driven R&D and universities responsible for education and research will collaborate. Leveraging the characteristics and strengths of both, we will

further exchanges among researchers, faculty, and students, and strengthen the development of research talent.

- ✓ Furthermore, we will utilize science museum exhibits, video content, and other resources to create environments where children can engage with quantum concepts from an early age, striving to cultivate a generation of quantum natives.

(iv) Other (Points to Note, etc.)

- ✓ Both industry and academia face serious challenges in developing and securing talent. Therefore, universities, research institutions, and others are expected to actively expand positions, while industry is expected to actively accept them, thereby working together to enhance and expand their respective systems.

(4) Intellectual Property and Standardization of Quantum Technologies

<Key Points>

- **Formation of a private-sector-led patent pool and establishment of a privately operated organization, incorporating open-close strategies**
- **Establishing frameworks and mechanisms to lead intellectual property protection and standardization of quantum technologies (e.g., quantum computers, quantum cryptography), and international rule-making**
- **Advancing practical technologies through quantum cryptography communication demonstrations, achieving pioneering intellectual property and standardization globally, and establishing practical technologies including peripheral technologies and their standardization**

(i) Formation of a private-sector-led patent pool and establishment of a privately operated organization, incorporating open-close strategies

- ✓ Quantum computers and quantum cryptography communications are integrated systems combining diverse technologies, including quantum and conventional (classical) technologies. It is anticipated that a complex web of diverse intellectual property and rights will emerge in the future.
- ✓ Going forward, to facilitate the smooth utilization of this intellectual property by private companies and others in their business activities and R&D, the formation of a private-sector-led international patent pool for bundled licensing of multiple intellectual property rights is anticipated. In this context, an open-close strategy—involving discussions and identification among stakeholders of patents related to quantum technology that should be secured and shared—is also crucial.
- ✓ Therefore, the government will promote such open-close strategies in collaboration with private companies and provide necessary support for the establishment of a private-sector-led patent pool and the organization to operate it.

(ii) Establishing systems and mechanisms to lead the intellectual property protection, standardization, and international rule-making for quantum technologies such as quantum computers and quantum cryptography

- ✓ Regarding the intellectual property protection of quantum technologies, Japan ranks third in the number of patents obtained, following China and the United States, and internationally...

Japan's level of intellectual property rights is relatively high. However, China and the United States have shown rapid growth over the past decade, and it is important for Japan to strengthen its efforts toward intellectual property rights. Regarding standardization, Japan actively participates not only in de jure standards bodies such as the International Telecommunication Union Telecommunication Standardization Sector (ITU-T) and the International Electrotechnical Commission (IEC), but also in forums for discussing de facto standards like the Institute of Electrical and Electronics Engineers (IEEE). Japan is expected to play a leading role in these areas going forward.

- ✓ Therefore, with an eye toward future computing and communications systems, we will continue to advance the intellectual property protection and standardization of quantum computers and quantum cryptography. This includes establishing a framework and mechanisms to lead international rule-making, encompassing the development of a unified public-private system for this purpose and support for private-sector standardization activities.
- ✓ Furthermore, each organization forming a quantum technology innovation hub will establish specialized groups to support intellectual property and standardization activities. Efforts will also be made to create incentives within universities and research institutions to promote intellectual property development.

(iii) Advancing practical technologies through quantum cryptography communication demonstrations, achieving world-leading intellectual property and standardization, and establishing practical technologies including peripheral technologies and their standardization.

- ✓ Regarding quantum technology, it is crucial to conduct user-driven proof-of-concept demonstrations as early as possible, in parallel with research and development. Insights gained from these demonstrations should be fed back into R&D to establish and advance practical technologies, driving intellectual property protection and standardization.
- ✓ Particularly for quantum cryptography communications, where practical implementation is advancing, we will leverage the open testbed for quantum cryptography communication networks. By enabling many users to conduct proof-of-concept testing, we will advance practical technologies and take the lead globally in intellectual property protection and standardization.
- ✓ Furthermore, beyond core component improvements such as extending range and miniaturization, we will establish practical technologies encompassing peripheral technologies (architecture, frameworks, interfaces, etc.), and advance intellectual property protection and standardization.

(iv) Other Considerations

- ✓ Since most quantum markets are in their infancy, it is important to consider open-close strategies according to the market's growth stage. This may involve utilizing public funding or government procurement when necessary, prioritizing social implementation through open innovation formats, and transitioning to closed systems once a certain market scale is achieved.

(5) International Collaboration / Industry-Academia-Government Collaboration

<Key Points>

- **Strengthening strategic international collaborative research, sending researchers—particularly young scientists—abroad, and attracting outstanding researchers from overseas to enhance international exchange and mobility**
- **Activating international exchange and cooperation within industry, and supporting industry's overseas expansion**
- **Establishing collaboration and cooperation frameworks between industry, quantum technology innovation hubs, and relevant government ministries (including forums for exchanging opinions, personnel exchanges, joint research, etc.)**

(i) Enhancing strategic international joint research, increasing international exchange and mobility through overseas dispatch of researchers (primarily young researchers) and attracting outstanding researchers from abroad

- ✓ In recent years, many researchers have pointed out the issue of declining international exchange of research personnel, particularly among young researchers. Furthermore, the impact of the COVID-19 pandemic has exacerbated this problem.
- ✓ Therefore, we will actively support the overseas dispatch of researchers, primarily young researchers, across a wide range of fields from basic to applied research. Simultaneously, we will enhance international exchange and mobility by attracting outstanding researchers from abroad.
- ✓ Furthermore, while promoting the establishment of frameworks for cooperation between nations, we will vigorously advance international joint research, led by quantum technology innovation hubs, in collaboration with overseas universities, research institutions, private companies, and others possessing cutting-edge technologies, to incorporate the latest technologies from abroad.

(ii) Activating international exchange and cooperation within industry and supporting industry's overseas expansion

- ✓ For private companies to develop and manufacture quantum computers and quantum cryptography communication systems, it is essential to establish international supply chains and collaborative R&D frameworks. To achieve this, international exchange and cooperation among industries is indispensable.
- ✓ Therefore, the government will actively promote and support international exchange and cooperation within industry, centered on the Quantum Technology-based New Industry Creation Council (Q-STAR), utilizing multilateral cooperation platforms as necessary.
- ✓ Furthermore, while identifying Japan's strengths that are globally competitive, the government and private sector will work together to promote the overseas expansion of products and services utilizing quantum technology.

(iii) Establishment of a collaboration and cooperation framework between industry, quantum technology innovation hubs, and relevant ministries and agencies (including forums for exchanging opinions, personnel exchanges, joint research, etc.)

- ✓ In February 2021, eight Quantum Innovation Hubs (QIH) were established at and . Concurrently, within industry, the Quantum Technology-based New Industry Creation Council (Q-STAR) was founded in September of the same year. These developments established new frameworks to advance the research, development, and industrialization of quantum

technologies.

- ✓ Additionally, the Quantum Innovation Initiative Council (QII Council), a collaboration between the University of Tokyo, Keio University, IBM USA, IBM Japan, and industry partners, was established in July 2020. Concurrently, the first commercial quantum computer was installed in Japan, and its use by industry-academia members commenced in July 2021.
- ✓ Going forward, to enable industry, academia, and government to work together on research and development, industrialization, and other initiatives, we will establish a framework for the Quantum Innovation Hub (QIH), the New Industry Creation Council (Q-STAR), and relevant ministries and agencies to collaborate and cooperate systematically. We will also leverage industry-academia collaboration facilitated by the Quantum Innovation Initiative Council (QII Council). Furthermore, under this framework of collaboration and cooperation, we will actively promote initiatives such as industry-academia-government discussions, joint research, personnel exchanges, and various events, responding to the mutual needs of stakeholders.

(6) Promotion of Outreach Activities

<Key Points>

- **Enhancing and strengthening public outreach activities through media content such as science museum exhibits, social media, and videos; creating environments where young people can engage with quantum technology**
- **Strengthening and enhancing information provision mechanisms (such as portal sites) that centrally provide information on quantum technology**
- **Promoting frontrunner talent such as quantum evangelists/ambassadors, entrepreneurs, and researchers who bridge quantum technology and society (also contributing to youth career development)**

(i) **Strengthening and enhancing public outreach activities through media and content such as science museum exhibits, social media, and videos; creating environments where young people can engage with quantum technology**

- ✓ Quantum technology is generally difficult to understand and has a high barrier to entry. Therefore, it is important to further enhance and strengthen public outreach activities to convey quantum technology in an accessible manner to a broad audience.
- ✓ Therefore, we will utilize media such as SNS and YouTube, along with video content, to clearly explain quantum technology, including its practical applications.
- ✓ Additionally, we will create environments where younger audiences, such as children, students, and young adults, can experience quantum technology through science museum exhibits and the development of hands-on educational materials.

(ii) **Enhancing and strengthening information provision mechanisms, such as a centralized portal site for quantum technology information.**

- ✓ Currently, information on the activities of the Quantum Innovation Hub (QIH) and the Quantum Technology New Industry Creation Council (Q-STAR), as well as educational programs provided by related organizations, is disseminated through individual websites and other channels.

- ✓ However, for users, the need to access each resource individually presents challenges of low convenience and difficulty in understanding. Furthermore, as the number of stakeholders continues to grow, there is a risk of increasing complexity.
- ✓ Considering these points, to provide information on quantum technology to society in an easily understandable form, we will enhance and strengthen information provision, including the creation of a portal site that centrally provides such information.

(iii) Connecting quantum technology with society through talent (evangelists, ambassadors, etc.), and promoting frontrunner talent such as entrepreneurs and researchers (also contributing to youth career development)

- ✓ Quantum technology is generally difficult for the public to understand. Therefore, mechanisms and individuals are needed to translate and communicate what quantum technology entails and what it can achieve in a way that society can easily grasp.
- ✓ Therefore, we will appoint individuals who bridge quantum technology and society (such as evangelists and ambassadors) to raise societal awareness using SNS, YouTube, and other media to raise societal awareness.
- ✓ Additionally, to contribute to the career development of young professionals, we will explore mechanisms (such as information provision and recognition programs) to highlight frontrunner talent—entrepreneurs, researchers, engineers, and others—who are creating and delivering new value through quantum technology.

(7) Economic Security / Business Environment Improvement, etc.

<Key Points>

- **Ensuring and strengthening economic security (acquiring cutting-edge technologies to secure technological superiority, securing supply chains for critical components and materials, responding to cyberattacks, etc.)**
- **Supply of risk capital through government-affiliated funds and promotion of demand stimulation by early adopters (government procurement)**
- **Improving the operation of national projects, such as utilizing fund systems enabling long-term investment**

(i) Ensuring and Strengthening Economic Security (Acquiring cutting-edge technologies to secure technological superiority, securing supply chains for critical foundational components and materials, responding to cyberattacks, etc.)

- ✓ We will strategically pursue the acquisition of cutting-edge technologies to secure technological superiority, including foundational materials, semiconductor devices, control and measurement equipment, peripheral devices, and manufacturing equipment necessary for quantum computers, quantum networks, and quantum measurement/sensing. We will also work to secure supply chains for critical foundational components and materials. Particularly for single-photon detectors, which are essential for quantum communication and currently rely heavily on overseas sources, we will work towards domestic production.
- ✓ Research and development elements related to quantum technology require the latest technologies. Beyond- , 5G Promotion Strategy,

Given the many commonalities with semiconductor strategies and others, we will strengthen collaboration to ensure research and development based on these strategies proceeds in an integrated and effective manner.

- ✓ Furthermore, we will take measures to protect critical information systems and data that form the foundation of industry and society, including technological information, from cyberattacks and other threats.

(ii) Supply of risk capital through government-affiliated funds and promotion of demand stimulation by early adopters (government procurement)

- ✓ Quantum computing and other quantum technologies face numerous technical challenges that must be resolved before practical application, requiring long-term investment.
- ✓ In Japan's private sector (including financial institutions), private investment tends to be sluggish when future markets are uncertain and investments span long periods. Therefore, to reduce risk and stimulate private investment, risk capital will be supplied through government-affiliated funds and similar mechanisms.
- ✓ Furthermore, when private companies develop and release new products or services (including prototypes) utilizing quantum technology, the government and relevant agencies will actively procure them as early adopters and strive to stimulate demand.

(iii) Improving the management of national projects, such as utilizing fund systems that enable long-term investment.

- ✓ Quantum technology requires long-term and challenging R&D investment. To strategically and sustainably advance R&D while considering the international competitive landscape, it is expected that improvements will be made to the operation of national projects. This includes utilizing fund systems such as the moonshot R&D system within national projects.

6. Finally

The 20th century, which saw the birth of quantum technology, witnessed the successive emergence of new technologies based on quantum mechanics—such as nuclear power, semiconductor electronics originating from the transistor, and lasers—which drove scientific and technological innovation and societal progress.

However, quantum mechanics encompasses principles unique to itself, such as the collapse of a wave function—where a superposition state collapses into a single value upon observation—and quantum entanglement. These represent a qualitative departure from the classical physics we are familiar with and became subjects for academic inquiry, such as the measurement problem in quantum theory. By the latter half of the 20th century, it came to be positioned as a dream technology capable of bringing about new technological innovations.

Since the start of this century, advances in both experimental and theoretical research have propelled quantum technology from "verification" to "control" at a rapid pace. For instance, in quantum computing, the once-distant dream of precisely controlling the quantum states of individual atoms and electrons for use in quantum computation is becoming a reality.

In the future, it will become possible to control larger-scale quantum systems. This will usher in an era where quantum technology is fully utilized for information processing, secure communications, and ultra-high-sensitivity sensing. This vision outlines the fundamental concepts and direction of efforts for effectively utilizing quantum technology in this new era.

These are technologies humanity has yet to master, and they are expected to transform the future socio-economic landscape and bring unprecedented benefits. Conversely, however, if used with malicious intent, they could cause immense harm. Therefore, we must proactively engage with these new technologies, grasping their very essence, and ensure they are harnessed as humanity's shared assets. This means not only pursuing economic growth but also striving for harmony between people and the environment, and aiming for human well-being—a life enriched by the heart. We should lead the innovation in this field.

The advancement of quantum technology to date is the result of researchers' curiosity and diligent efforts. Going forward, industry will play a major role in delivering the benefits of quantum technology to society.

We look forward to industry and academia joining hands to further advance quantum technology and bring significant benefits to the socio-economic sphere.